Duke University
Math 106L: Laboratory Calculus and Functions II

Exam 2

The following instructions will appear on the exam:

- Do not open this test booklet until you are directed to do so.
- You will have 105 minutes to complete the exam. If you finish early go back and check your work.
- This exam is closed book. However, you may use one side of a letter-sized piece of paper with your own *hand written* notes.
- You may not use a calculator or any other technology.
- Throughout the exam, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.

Note that Rann Bar-On's section did not cover the material for question 5.

Question 1 (12 points)

(a) Write out rules and bounds for the following sequences:

(i) 1, 1.6, 2.2, 2.8, 3.4, 4: $a_k =$ _____, ___ $\leq k \leq$ ___

(ii) 2, 4, 8, ..., 128: $b_k =$ _____, ___ $\leq k \leq$ ___

(b) Write the following sums with the given bound:

(i) $\displaystyle\sum_{k=0}^{20} \frac{1}{k^2+2} = \overline{\sum_{k=-5}} _____$

(ii) $\displaystyle\sum_{k=0}^{n-1} a_k = \overline{\sum_{k=___}} \; a_{n-k}$

Question 2 (9 points)

(a) Fill in the blanks: The Fundamental Theorem of Calculus, Part I (a.k.a. FTC I), states that if $f(x)$ is _____ on the interval $[a, b]$, and $F(x)$ is an _____ of $f(x)$, then $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$.

To prove this, we showed that repeated _____ _____ of $F(x)$, starting from $(a, F(a))$ is the same as the _____ _____ sum approximation of the area "under" _____ between $x = a$ and $x = b$.

(b) Why does FTC I not help with the following integrals?

(i) $\displaystyle\int_0^4 \frac{1}{x-2}\,dx$

(ii) $\displaystyle\int_0^4 e^{-x^2}\,dx$

Question 3 (17 points) Consider the sequence

$$a_k = \frac{2k}{5}\left(\frac{2k}{5} - 2\right)\left(\frac{2k}{5} - 4\right)$$

(a) Fill in the blanks: If $f(x) = x(x-2)(x-4)$, the sum $\displaystyle\sum_{k=0}^{4} a_k \cdot \frac{2}{5}$ is a _____-hand sum for $\displaystyle\int_{_}^{2} f(x)\,dx$ with _____ subintervals and $\Delta x =$ _____. (Don't forget to fill in the lower bound of the integral!)

(b) Write a midpoint sum with the same number of intervals for the same integral.

(c) By drawing the graph of $f(x)$ using your knowledge of polynomials, taking derivatives, or otherwise, explain why the midpoint sum is always an overestimate of this integral.

(d) Why is it true that for this integral, and for every n, $\text{LHS}_n = \text{RHS}_n$? (Hint: What do you know about $|\text{LHS}_n - \text{RHS}_n|$?)

(e) Use part (d) to decide whether the sum in part (a) of this question is an overestimate, underestimate, or the exact value of the integral. Explain your answer. (Hint: consider the average of the left hand and right hand sums.)

Question 4 (20 points) Compute the following integrals.

(a) $\displaystyle\int \sin\left(\frac{x}{17}\right) + e^{3x} + \sqrt{x} + 2 \; dx$

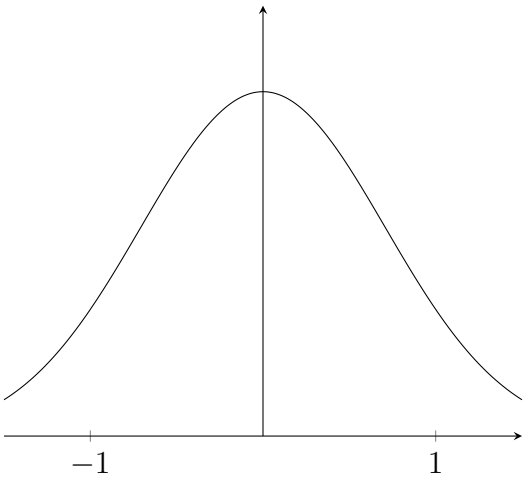
(b) $\displaystyle\int \arcsin x \; dx$

(c) $\displaystyle\int_0^2 x e^{x^2} \; dx$

(d) $\displaystyle\int_0^2 x^2 e^x \; dx$

Question 5 (4 points) Find a function satifying the initial value problem $f'(x) = \cos(x^2)$, with $f(2) = 6$. (Hint: don't try to find a nice antiderivative!)

Question 6 (8 points) This question concerns the integral $\int_{-1}^{1} e^{-x^2}\, dx$. The graph of e^{-x^2} is shown below:



(a) Show that the integral is less than 2, but greater than $\frac{2}{e}$. (Hint: Feel free to draw on and label the graph above!)

(b) Given that $1 - \frac{x^2}{2} \geq e^{-x^2} \geq 1 - x^2$ on $[-1, 1]$, find better upper and lower bounds for the value of the integral than the ones given in part (a).

Question 7 (15 points) You drop some dye into a cylinder-shaped container of water, with radius 5 cm and height 10 cm. The dye is denser than water, so it begins sinking. A little while after you poured the dye into the water, you measure the density of dye throughout the cylinder and find that it is closely modeled by the function

$$D(h) = 10e^{0.1h} \text{ grams of dye per cm}^3 \text{ of water,}$$

where h is the distance from the bottom of the cylinder.

(a) How should you divide the cylinder up in order to estimate the total amount of dye? Explain.

(b) If you divide the cylinder up into five subdivisions, find an overestimate and an underestimate for the total amount of dye in the second subdivision from the bottom of the cylinder. Leave your answer in terms of e.

(c) Write down a Riemann sum estimating the total amount of dye in the cylinder, using n subdivisions. (You can do an LHS, RHS, or MPS. Up to you.)

(d) By finding and evaluating the appropriate definite integral, find the exact total amount of dye in the cylinder. Again, leave your answer in terms of e.

Question 8 (15 points) Decide whether each of the following statements is true or false. If it is false, provide a reason or counterexample.

(a) $G(x) = \ln|x^2|$ is an antiderivative of $g(x) = \frac{1}{x^2}$.

(b) The average velocity of a car can be calculated by finding the area under the velocity curve and dividing by the total time the journey takes.

(c) If a function $f(x)$ is decreasing on $[a, b]$, then the midpoint sum always underestimates $\int_a^b f(x)\,dx$.

(d) $\dfrac{d}{dx}\left(\displaystyle\int_0^2 \ln(\cos(\sqrt{x}))\,dx\right) = \ln(\cos(\sqrt{x}))$.